What negative Yamabe normalization preserves—and what it does not

23 September 2026. Pointwise almost-Einstein input admits a uniformly small scalar-normalizing conformal change in C^1 and normalized $W^{2,2}$, without injectivity-radius, Sobolev, or scalar spectral-gap assumptions. Nevertheless that conformal change need not preserve pointwise Ricci approximation: the normalized Ricci error can diverge. An explicit counterexample below occurs on a fixed product of hyperbolic surfaces, with bounded input sectional curvature and no collapse.

Throughout, $\Delta = d^*d = -\operatorname{div}\nabla$, and all norms of u and its derivatives use the original metric unless indicated.

1. A uniform positive result

Theorem 1. Let (M^4, g) be closed, connected, and smooth, and suppose

$$\|g^{-1}\operatorname{Ric}_g + I\|_{\infty,\mathrm{op}} \leqslant \varepsilon \leqslant \frac{1}{4}. \tag{1}$$

There is a unique positive smooth function u for which $\widehat{g} = u^2 g$ has scalar curvature -4. Put $e = s_g + 4$, $V = \operatorname{Vol}(g)$. Then

$$1 - \varepsilon \leqslant u^2 \leqslant 1 + \varepsilon, \qquad \|du\|_\infty \leqslant 4\varepsilon, \tag{2}$$

In particular $\|g^{-1}\widehat{g} - I\|_\infty \leqslant \varepsilon$ and $(1-\varepsilon)^2 V \leqslant \operatorname{Vol}(\widehat{g}) \leqslant (1+\varepsilon)^2 V$.

$$\int \left((u-1)^2 + |du|^2 + |\nabla^2 u|^2\right) \leqslant \frac{3}{8}\int e^2 \leqslant 6\varepsilon^2 V. \tag{3}$$

The normalized metric has the integral Einstein-error bound

$$\int |\operatorname{Ric}_{\widehat{g}} + \widehat{g}|^2_{\widehat{g}} \leqslant \int |\operatorname{Ric}^\circ_g|^2_g \leqslant 4\varepsilon^2 V, \qquad \fint_{\widehat{g}} |\operatorname{Ric}_{\widehat{g}} + \widehat{g}|^2 \leqslant \frac{4\varepsilon^2}{(1-\varepsilon)^2}. \tag{4}$$

Thus the scalar-normalizing change is C^1-small, with a second-order integral estimate. No bound on $\nabla\operatorname{Ric}$ or full curvature has been assumed. No monopole assumption is needed.

Proof of existence, amplitude, and integral estimates

Since $s_g \leqslant -3$, the conformal Yamabe constant is negative. The Yamabe theorem gives a negative constant-scalar representative, which can be uniquely rescaled to scalar curvature -4. Uniqueness also follows directly: the ratio of two such factors solves $6\Delta z - 4z + 4z^3 = 0$ on one normalized metric; the maximum and minimum principles force $z = 1$.

The factor satisfies

$$6\Delta u + (-4 + e)u + 4u^3 = 0. \tag{5}$$

At its maximum and minimum the signs of Δu give (2)'s amplitude bound. Write $v = u - 1$ and $a = 4u(u+1)$. Then

$$6\Delta v + av = -eu, \qquad 6 \leqslant a \leqslant 10, \qquad u \leqslant 9/8. \tag{6}$$

Pairing with v gives

$$\|v\|_2 \leqslant \frac{3}{16}\|e\|_2, \quad \|dv\|_2^2 \leqslant \frac{9}{256}\|e\|_2^2, \quad \|\Delta v\|_2 \leqslant \frac{1}{2}\|e\|_2. \tag{7}$$

The last bound follows by substituting the first one into (6). Integrated Bochner and $\mathrm{Ric} \geqslant -5g/4$ give

$$\|\nabla^2 v\|_2^2 = \|\Delta v\|_2^2 - \int \mathrm{Ric}(dv, dv) \leqslant \frac{301}{1024}\|e\|_2^2. \tag{8}$$

Adding the three bounds proves (3). These are estimates on the original closed metric, not elliptic estimates in uniformly controlled charts.

Proof of the maximum gradient bound

Rewrite (5) as

$$(\Delta + 4/3)v = F, \qquad F = -eu/6 - 2v^2 - (2/3)v^3, \qquad \|F\|_\infty \leqslant 2\varepsilon. \tag{9}$$

Here $|v| \leqslant \varepsilon$ and $|e| \leqslant 4\varepsilon$. For the heat semigroup $P_t = e^{-t\Delta}$, Bochner under $\mathrm{Ric} \geqslant -Kg$ gives

$$|\nabla P_t f|^2 \leqslant \frac{K}{1 - e^{-2Kt}}\left(P_t(f^2) - (P_t f)^2\right) \leqslant \left(K + \frac{1}{2t}\right)\|f\|_\infty^2. \tag{10}$$

One direct derivation is to differentiate $P_s[(P_{t-s}f)^2]$, integrate from 0 to t, and use $|\nabla P_t f|^2 \leqslant e^{2Ks}P_s|\nabla P_{t-s}f|^2$. The second inequality uses $1 - e^{-x} \geqslant x/(1+x)$. The positive-mass resolvent therefore satisfies

$$\|\nabla(\Delta + m)^{-1} f\|_\infty \leqslant \left(\frac{\sqrt{K}}{m} + \sqrt{\frac{\pi}{2m}}\right)\|f\|_\infty. \tag{11}$$

Indeed, integrate $e^{-mt}\nabla P_t f$ over $t > 0$; the bound is integrable both at zero and at infinity. With $K = 5/4$ and $m = 4/3$, the coefficient is $3\sqrt{5}/8 + \sqrt{3\pi/8} < 2$. Equations (9)–(11) prove (2).

Proof of the Ricci-energy bound

For a negative Yamabe class, $\int s_{\widehat{g}}^2 \leqslant \int s_g^2$. For example, if $\widehat{g}_0 = z^2 g$ has volume one, its Yamabe constant is $Y = \int (6|dz|^2 + s_g z^2) \geqslant -\|s_-\|_2$; hence $\int s_{\widehat{g}}^2 = Y^2 \leqslant \int s_g^2$. In four dimensions, $\int |W^+|^2$ is conformally invariant and

$$\int |\mathrm{Ric}^\circ|^2 = 4\int |W^+|^2 + \frac{1}{12}\int s^2 - 8\pi^2(2\chi + 3\tau). \tag{12}$$

This is the conformal monotonicity proved in `line1_energy_selection.md`, Section 4; the argument is included here to make its lack of monopole assumptions explicit. Thus the trace-free Ricci energy cannot increase. Since $s_{\widehat{g}} = -4$, this is its full Einstein-error energy. Finally $\mathrm{Vol}(\widehat{g}) \geqslant (1-\varepsilon)^2 V$. This proves (4). All integrals and averages on the left side of (4) use $d\mu_{\widehat{g}}$, not the input volume measure. For a nonorientable manifold, apply the curvature-energy argument on its oriented double cover and divide the integrals by two; uniqueness makes the scalar-normalizing factor descend. The same estimates therefore hold.

2. An actual failure of pointwise Ricci preservation

Theorem 2. There is a fixed closed manifold $M = \Sigma_1 \times \Sigma_2$, with both genera at least two, and smooth metrics g_L such that

$$\|g_L^{-1}\,\mathrm{Ric}_{g_L} + I\|_\infty = O((\log L)/L) \to 0, \tag{13}$$

while their scalar-−4 Yamabe representatives satisfy

$$\liminf_{L\to\infty} \frac{1}{\log L}\|\widehat{g}_L^{-1}\,\mathrm{Ric}_{\widehat{g}_L} + I\|_\infty \geqslant \frac{2}{3}. \tag{14}$$

In particular, the normalized maximum Ricci error tends to infinity. The input metrics have uniformly bounded sectional curvature, diameter, and positive injectivity radius and scalar spectral gap. They converge in $C^{1,\alpha}$, for every fixed $0 < \alpha < 1$, to a fixed negative Einstein product. Thus none of those geometric controls repairs the failed endpoint conclusion.

A smooth two-dimensional bump

Fix hyperbolic metrics h_i of curvature -1. In an isothermal coordinate disk about $p \in \Sigma_1$, choose coordinates with $h_1(p) = dx^2 + dy^2$. Take a fixed radial smooth cutoff χ, equal to one near $p = (0,0)$, supported inside the disk. Put

$$v_L(x,y) = -\frac{\log L}{2L}\,\chi(x,y)\,xy\log(x^2+y^2+e^{-2L}), \tag{15}$$

for $L > e$, and extend by zero. With $r^2 = x^2 + y^2$ and $b > 0$, direct differentiation gives

$$(\partial_x^2 + \partial_y^2)[xy\log(r^2+b)] = 4xy\frac{2r^2+3b}{(r^2+b)^2}. \tag{16}$$

Its absolute value is at most four. The function and its first derivatives are bounded uniformly for $b = e^{-2L}$; cutoff errors have fixed support away from the singular point. In two dimensions the isothermal Laplacian is a smooth fixed multiple of the Euclidean one. Consequently

$$\|v_L\|_{C^1} + \|\Delta_{h_1} v_L\|_\infty = O((\log L)/L), \quad \|\nabla^2_{h_1} v_L\|_\infty \leqslant C\log L, \quad \nabla^2_{h_1} v_L(p) = (\log L)H, \tag{17}$$

where $H = dx\otimes dy + dy\otimes dx$. At p, $dv_L = 0$, so no connection term changes this Hessian. Interpolation gives

$$[dv_L]_{C^{0,\alpha}} \leqslant C_\alpha \|dv_L\|_\infty^{1-\alpha}\|\nabla^2 v_L\|_\infty^{\alpha} \leqslant C_\alpha(\log L)L^{\alpha-1} \longrightarrow 0$$

for every fixed $\alpha < 1$. Thus $v_L \to 0$ in $C^{1,\alpha}$ despite the unbounded central Hessian.

The input Ricci tensor

Set

$$g_L = e^{2v_L}h_1 + h_2. \tag{18}$$

The first factor's Gaussian curvature is $K_L = e^{-2v_L}(-1 + \Delta_{h_1} v_L)$; the second has curvature -1. Thus (17) proves (13). All mixed sectional curvatures vanish, and the two factor curvatures remain in a fixed negative interval. This gives uniform sectional-curvature bounds. Bilipschitz convergence to the fixed product bounds diameter and the scalar spectral gap. There are no conjugate points on either factor, and lengths of noncontractible loops are bounded below by bilipschitz comparison; the product therefore also has a uniform positive injectivity radius.

Exact scalar normalization exposes the Hessian

Seek a positive scalar-normalizing factor depending only on Σ_1. Its equation, after multiplication by e^{2v}, is

$$6\Delta_{h_1} u + (-2 + 2\Delta_{h_1} v - 2e^{2v})u + 4e^{2v}u^3 = 0. \tag{19}$$

The substitution $u = e^{-v/3}w$ cancels the entire Δv term and gives

$$6\Delta_{h_1} w + 4\langle dv, dw\rangle + \left(-\frac{2}{3}|dv|^2 - 2 - 2e^{2v}\right)w + 4e^{4v/3}w^3 = 0. \tag{20}$$

For fixed $0 < \alpha < 1$, the left side is a smooth map from $C^{1,\alpha}(\Sigma_1)\times C^{2,\alpha}(\Sigma_1)$ to $C^{0,\alpha}(\Sigma_1)$. At $(v, w) = (0, 1)$, its derivative in w is $6\Delta_{h_1} + 8$, an isomorphism. The implicit function theorem supplies positive $w_L \to 1$ in $C^{2,\alpha}$. Then $u_L = e^{-v_L/3} w_L$ solves the actual four-dimensional scalar equation. Uniqueness in Theorem 1 identifies it with the negative Yamabe factor; no symmetry of a hypothetical factor was assumed.

At p, (17) now gives

$$u_L(p) = w_L(p) \to 1, \quad du_L(p) = dw_L(p) \to 0, \quad \nabla^2_{g_L} u_L(p) = \nabla^2_{h_1} w_L(p) - \frac{w_L(p)\log L}{3} H. \tag{21}$$

Here the connections of g_L and the fixed product converge in C^0, and $dv_L(p) = 0$. The four-dimensional conformal formula is

$$\mathrm{Ric}^{\circ}_{u^2 g} = \mathrm{Ric}^{\circ}_{g} - 2u^{-1}(\nabla^2_g u)^{\circ} + 4u^{-2}(du \otimes du)^{\circ}. \tag{22}$$

The trace-free operation on covariant two-tensors is unchanged by a conformal rescaling. At p, the background connection is precisely the fixed product connection because $v_L(p) = dv_L(p) = 0$. Also $\Delta v_L(p) = 0$, so the input Ricci tensor is exactly $-g_L$ there. The first term on the right of (21), extended by zero on the second factor, tends to zero. Substitution into (22) therefore gives the covariant-tensor expansion

$$(\mathrm{Ric}_{\widehat{g}_L} + \widehat{g}_L)(p, y) = \frac{2}{3}(\log L) H + o(1).$$

The coefficient $w_L(p)$ cancels against $u_L(p)^{-1}$. Raising an index uses $\widehat{g}_L(p, y) = w_L(p)^2 g_0(p, y)$. Since $w_L(p) \to 1$ and $\|H\|_{\mathrm{op}} = 1$, the pointwise operator norm divided by $\log L$ tends to $2/3$. This proves (14). If the prefactor $\log L$ is replaced by one throughout, the same argument gives the finite endpoint defect $2/3$; the slowly growing prefactor establishes the stronger unbounded statement.

3. Consequence for the first research line

The counterexample is compatible with all integral results above: $\|\nabla^2 u_L\|_2 \to 0$, even though its value at p diverges. Each input is a Kähler product on a fixed general-type surface. For its canonical class the exact Kähler identities give $D_{LB} = \int (s - \bar{s})^2/(64\pi^2) \to 0$ and $p - a^2 \to 0$. Thus the obstruction is not missing monopole data, negative-square control, or a scalar spectral gap.

This rules out one proposed passage from pointwise input to a convenient scalar-normalized pointwise input. It does not rule out other corrections: the limiting product itself is Einstein. The theorem also does not show that actual weighted-monopole forms must have zeros; that is a different question and requires its own argument.

Round 14 normalized a specific shear family whose lifted coefficients had uniform higher-derivative bounds, sufficient for Schauder estimates. Theorem 1 needs no such bounds, and Theorem 2 explains why its universal conclusion must stop short of a maximum-norm Hessian or Ricci estimate. The logarithmic bump is the classical endpoint Laplacian/Hessian mechanism; the contribution here is its realization inside an actual Ricci-pinched family and its exact scalar-normalizing equation.

`line1_yamabe_endpoint_check.py` verifies finite differential identities, norm conventions, and constants. Existence uses the classical negative Yamabe theorem and the explicitly stated fixed-background implicit function theorem, not the finite checks.